

DETERMINANTS

WORKEDOUT EXAMPLES

W.E-1: If $\begin{vmatrix} x & 3 & 6 \\ 3 & 6 & x \\ 6 & x & 3 \end{vmatrix} = \begin{vmatrix} 2 & x & 7 \\ x & 7 & 2 \\ 7 & 2 & x \end{vmatrix} = \begin{vmatrix} 4 & 5 & x \\ 5 & x & 3 \\ x & 4 & 5 \end{vmatrix} = 0$

then x is equal to

- A)9 B)-9
C)0 D)None of these

Sol: (B)

By circulant determinant property

$$a + b + c = 0$$

$$x + 3 + 6 = x + 2 + 7 = x + 4 + 5 = 0$$

$$x = -9$$

W.E-2: If $f(x)$ is a polynomial of degree < 2 , prove that

$$\begin{vmatrix} 1 & a & \frac{f(a)}{x-a} \\ 1 & b & \frac{f(b)}{x-b} \\ 1 & c & \frac{f(c)}{x-c} \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \frac{f(x)}{(x-a)(x-b)(x-c)}$$

Sol: Let

$$\frac{f(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c} \dots (1)$$

$$\Rightarrow f(x) = A(x-b)(x-c) + B(x-a)(x-c) + C(x-a)(x-b) \dots (2)$$

Put $x=a, b$ and c in eqn (2), we get

$$A = \frac{f(a)}{(x-b)(x-c)}, B = -\frac{f(b)}{(a-b)(b-c)} \text{ and}$$

$$C = -\frac{f(c)}{(b-c)(c-a)}$$

$$\therefore \frac{f(x)}{(x-a)(x-b)(x-c)} = \frac{\frac{f(a)(c-b)}{x-a} - \frac{(c-a)f(b)}{x-b} + \frac{(b-c)f(c)}{x-c}}{(a-b)(b-c)(c-a)}$$

$$= \begin{vmatrix} 1 & a & \frac{f(a)}{x-a} \\ 1 & b & \frac{f(b)}{x-b} \\ 1 & c & \frac{f(c)}{x-c} \end{vmatrix} \div \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 0$$

W.E-3: If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors, then show that

$$\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix} = 0$$

Sol: Given $\vec{a}, \vec{b}, \vec{c}$ are coplanar, there exist scalars k, l, m not all zero, say $k \neq 0$, such that $k\vec{a} + l\vec{b} + m\vec{c} = \vec{0}$

$$\text{Now } \Delta = \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix}$$

Operating $C_1 \rightarrow kC_1 + lC_2 + mC_3$ we get

$$\Delta = \frac{1}{k} \begin{vmatrix} k\vec{a} + l\vec{b} + m\vec{c} & \vec{b} & \vec{c} \\ k\vec{a} \cdot \vec{a} + l\vec{a} \cdot \vec{b} + m\vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ k\vec{b} \cdot \vec{a} + l\vec{b} \cdot \vec{b} + m\vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix}$$

$$\Delta = \frac{1}{k} \begin{vmatrix} k\vec{a} + l\vec{b} + m\vec{c} & \vec{b} & \vec{c} \\ \vec{a}(k\vec{a} + l\vec{b} + m\vec{c}) & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b}(k\vec{a} + l\vec{b} + m\vec{c}) & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix}$$

$$\Delta = \frac{1}{k} \begin{vmatrix} \vec{0} & \vec{b} & \vec{c} \\ \vec{0} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{0} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \end{vmatrix} = 0$$

W.E-4:

Let

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$$\Delta = \begin{vmatrix} \sin x & \sin(x+h) & \sin(x+2h) \\ \sin(x+2h) & \sin x & \sin(x+h) \\ \sin(x+h) & \sin(x+2h) & \sin x \end{vmatrix}$$

$$\text{Find } \lim_{h \rightarrow 0} \left(\frac{\Delta}{h^2} \right)$$

Sol: Apply

$$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_2$$

$$\lim_{h \rightarrow 0} \frac{\left(\frac{2\sin \frac{h}{2} \right)^2}{h^2} \begin{vmatrix} \sin x & \cos\left(x+\frac{h}{2}\right) & \cos(x+h) \\ \sin(x+2h) & -2\cos(x+h)\cos\frac{h}{2} & \cos\left(x+\frac{h}{2}\right) \\ \sin(x+h) & \cos\left(x+\frac{3h}{2}\right) & -2\cos(x+h)\cos\frac{h}{2} \end{vmatrix}$$

$$= \sin x \cos^2 x \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{vmatrix}$$

$$= 9 \sin x \cos^2 x$$

W.E-5: If a,b,c,d are positive and different, show that

$$\begin{vmatrix} 1 & bc+ad & b^3c^3+a^3d^3 \\ 1 & ca+ad & c^3a^3+b^3d^3 \\ 1 & ab+cd & a^3b^3+c^3d^3 \end{vmatrix} \neq 0$$

Sol: Let $bc+ad = \alpha, ca+bd = \beta, ab+cd = \gamma$

Then given determinant =

$$\begin{vmatrix} 1 & \alpha & \alpha^3-3abcd\alpha \\ 1 & \beta & \beta^3-3abcd\beta \\ 1 & \gamma & \gamma^3-3abcd\gamma \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \alpha & \alpha^3 \\ 1 & \beta & \beta^3 \\ 1 & \gamma & \gamma^3 \end{vmatrix} - 3abcd \begin{vmatrix} 1 & \alpha & \alpha \\ 1 & \beta & \beta \\ 1 & \gamma & \gamma \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \alpha & \alpha^3 \\ 1 & \beta & \beta^3 \\ 1 & \gamma & \gamma^3 \end{vmatrix} - 0$$

$$= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\alpha + \beta + \gamma)$$

$$= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$

$$(bc+ad+ca+bd+ab+cd) \neq 0$$

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because a,b,c,d are positive and different

W.E-6: If $f(x), g(x), h(x)$ are polynomials of degree 4 and

$$\begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ p & q & r \end{vmatrix} = mx^4 + nx^3 + rx^2 + 5x + t$$

be an identity in x, then show that

$$\begin{vmatrix} f''(0)-f'(0) & g''(0)-g'(0) & h''(0)-h'(0) \\ a & b & c \\ p & q & r \end{vmatrix} = 2(3n-r)$$

Sol:

Differentiating given equation w.r.t. x we get

$$\begin{vmatrix} f'(x) & g'(x) & h'(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 4mx^3 + 3nx^2 + 2rx + 5 \dots (1)$$

Again differentiating above equation, w.r.t. x we get

$$\begin{vmatrix} f''(x) & g''(x) & h''(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 12mx^2 + 6nx + 2r \dots (2)$$

Again differentiating w.r.t. x we get

$$\begin{vmatrix} f'''(x) & g'''(x) & h'''(x) \\ a & b & c \\ p & q & r \end{vmatrix} = 24mx + 6n \dots (3)$$

Putting $x=0$ in (3), we get

$$2r = \begin{vmatrix} f''(0) & g''(0) & h''(0) \\ a & b & c \\ p & q & r \end{vmatrix} \dots (4)$$

Putting $x=0$ in (3) we get

$$6n = \begin{vmatrix} f''(0) & g''(0) & h''(0) \\ a & b & c \\ p & q & r \end{vmatrix} \dots (5)$$

Subtracting (4) from (5), we get

$$\begin{vmatrix} f'''(0)-f''(0) & g'''(0)-g''(0) & h'''(0)-h''(0) \\ a & b & c \\ p & q & r \end{vmatrix}$$

$$= 6n - 2r = 2(3n - r)$$

W.E-7:

If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$

$$(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$$

$$(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$$

then prove that

$$4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c)$$

Sol:

area of the triangle PQR with vertices

 $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) is

$$\therefore \Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \dots\dots\dots(1)$$

Now Area of Δ PQR with sides a, b, c is

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{\frac{1}{16}(2s)(2s-2a)(2s-2b)(2s-2c)} \dots\dots\dots(2)$$

$$\Delta = \sqrt{\left[\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{16} \right]}$$

Squaring on both sides, we have

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c)$$

W.E-8: If $f(x) = \begin{vmatrix} x+c_1 & x+a & x+a \\ x+b & x+c_2 & x+a \\ x+b & x+b & x+c_3 \end{vmatrix}$ then show that

 $f(x)$ is linear in x

hence deduce that $f(0) = \frac{bg(a) - ag(b)}{(b-a)}$

here $g(x) = (c_1 - x)(c_2 - x)(c_3 - x)$

Sol.

$$f(x) = \begin{vmatrix} x+c_1 & x+a & x+a \\ x+b & x+c_2 & x+a \\ x+b & x+b & x+c_3 \end{vmatrix}$$

$$= \begin{vmatrix} x+c_1 & a-c_1 & 0 \\ x+b & c_2-b & a-c_2 \\ x+b & 0 & c_3-b \end{vmatrix} \quad C_2 \rightarrow C_2 - C_1; C_3 \rightarrow C_3 - C_2$$

$$f(x) = x \begin{vmatrix} 1 & a-c_1 & 0 \\ 1 & c_2-b & a-c_2 \\ 1 & 0 & c_3-b \end{vmatrix} + \begin{vmatrix} c_1 & a-c_1 & 0 \\ b & c_2-b & a-c_2 \\ b & 0 & c_3-b \end{vmatrix} \dots\dots(1)$$

so, $f(x)$ is linear.

$$\text{Let } f(x) = Px + Q$$

$$f(-a) = -aP + Q, \quad f(-b) = -bP + Q$$

$$f(0) = 0.P + Q = Q$$

$$= \frac{bf(-a) - af(-b)}{(b-a)} \dots\dots\dots(2)$$

from (1)

$$f(-a) = \begin{vmatrix} c_1 - a & 0 & 0 \\ b - a & c_2 - a & 0 \\ b - a & 0 & c_3 - a \end{vmatrix}$$

$$= (c_1 - a)(c_2 - a)(c_3 - a)$$

$$f(-b) = (c_1 - b)(c_2 - b)(c_3 - b)$$

$$g(x) = (c_1 - x)(c_2 - x)(c_3 - x)$$

$$g(a) = f(-a)$$

$$g(b) = f(-b)$$

Now from (2) we get

$$f(0) = \frac{bg(a) - ag(b)}{(b-a)}$$

W.E-9: Given $A = \begin{vmatrix} a & b & 2c \\ d & e & 2f \\ l & m & 2n \end{vmatrix}, B = \begin{vmatrix} f & 2d & e \\ n & 4l & 2m \\ c & 2a & b \end{vmatrix}$ then

the value of B/A is

Sol: (2) Taking 2 common from R_2 and C_2

We get

 $B = 2A$ with the help of

$$R_3 \leftrightarrow R_2 \text{ then } R_2 \leftrightarrow R_1$$

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W.E-10: If $S_k = a^k + b^k + c^k$ then $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} =$

$$\begin{vmatrix} 3 & a+b+c & a^2+b^2+c^2 \\ a+b+c & a^2+b^2+c^2 & a^3+b^3+c^3 \\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^4+b^4+c^4 \end{vmatrix} =$$

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}^2 = (a-b)^2 (b-c)^2 (c-a)^2$$

If we can express a determinant as a product of two determinants, then it is easier for us.

On the basis of above information, answer the following questions.

- i) a, b, c and d are complex numbers, then the value of

$$\begin{vmatrix} 2 & a+b+c+d & ac+bd \\ a+b+c+d & 2(a+c)(b+d) & ac(b+d)+bd(a+c) \\ ac+bd & ac(b+d)+bd(a+c) & 2abcd \end{vmatrix} \text{ is}$$

- (A) 0 (B) a+b+c+d
(C) abcd (D) ac+bd

- ii) If A, B, C, P, Q, R $\in \mathbb{R}$, then the value of

$$\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix} \text{ is}$$

- (A) $\cos A \cos P + \cos B \cos Q + \cos C \cos R$
(B) $\cos(A-P) \cos(B-Q) \cos(C-R)$
(C) $\cos(A+B+C) - \cos(P+Q+R)$
(D) none of the above

- iii) If $a_1, a_2, a_3, b_1, b_2, b_3 \in \mathbb{R}$, then the value of

$$\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 \end{vmatrix} \text{ is}$$

- (A) $2(a_1-a_2)(a_2-a_3)(a_3-a_1)$
(B) $2(b_1-b_2)(b_2-b_3)(b_3-b_1)$
(C) $2(a_1-a_2)(a_2-a_3)(a_3-a_1) \times (b_1-b_2)(b_2-b_3)(b_3-b_1)$
(D) 0

key: (A, D, C)

Sol: $\begin{vmatrix} 2 & a+b+c+d & ac+bd \\ a+b+c+d & 2(a+c)(b+d) & ac(b+d)+bd(a+c) \\ ac+bd & ac(b+d)+bd(a+c) & 2abcd \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 0 \\ b+d & a+c & 0 \\ bd & ac & 0 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 0 \\ a+c & b+d & 0 \\ ac & bd & 0 \end{vmatrix}$$

$$= 0 \times 0 = 0$$

$$\begin{vmatrix} \cos(A-P) & \cos(A-Q) & \cos(A-R) \\ \cos(B-P) & \cos(B-Q) & \cos(B-R) \\ \cos(C-P) & \cos(C-Q) & \cos(C-R) \end{vmatrix}$$

$$= \begin{vmatrix} \cos A & \sin A & 0 \\ \cos B & \sin B & 0 \\ \cos C & \sin C & 0 \end{vmatrix} \times \begin{vmatrix} \cos P & \sin P & 0 \\ \cos Q & \sin Q & 0 \\ \cos R & \sin R & 0 \end{vmatrix}$$

$$= 0 \times 0 = 0$$

$$\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 \end{vmatrix}$$

$$= \begin{vmatrix} a_1^2 & -2a_1 & 1 \\ a_2^2 & -2a_2 & 1 \\ a_3^2 & -2a_3 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & b_1 & b_1^2 \\ 1 & b_2 & b_2^2 \\ 1 & b_3 & b_3^2 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{vmatrix} \times \begin{vmatrix} 1 & b_1 & b_1^2 \\ 1 & b_2 & b_2^2 \\ 1 & b_3 & b_3^2 \end{vmatrix}$$

$$= 2(a_1-a_2)(a_2-a_3)(a_3-a_1)(b_1-b_2)(b_2-b_3)(b_3-b_1)$$

LEVEL - V

SINGLE ANSWER QUESTIONS

1. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are the

given determinants then

(A) $\Delta_1 = 3(\Delta_2)^2$ (B) $\frac{d}{dx} \Delta_1 = 3\Delta_2$

(C) $\frac{d}{dx} \Delta_1 = 3(\Delta_2)^2$ (D) $\Delta_1 = 3\Delta_2^{3/2}$

2. If $b^2 - ac < 0$, $a > 0$, $x \neq 0$, $y \neq 0$ then

$$\begin{vmatrix} a & b & ax+by \\ b & c & bx+cy \\ ax+by & bx+cy & 0 \end{vmatrix} \text{ is}$$

- A) >0 B) <0 C) $=0$ D) $b^2 + ac$

3. In a ΔABC , the value of

$$\begin{vmatrix} e^{-i(2A)} & e^{iC} & e^{iB} \\ e^{iC} & e^{-i2B} & e^{iA} \\ e^{iB} & e^{iA} & e^{-i2C} \end{vmatrix} \text{ is (where } i = \sqrt{-1} \text{)}$$

- (A) -2 (B) -4 (C) -3 (D) -1

4. If the value of the determinant $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix}$ is positive then

- (A) $abc < -2$ (B) $abc > 0$
(C) $abc > -8$ (D) $abc = 0$

5. If A, B, C are the angles of a triangle and a, b, c are the corresponding sides, then the value of the determinant

$$\begin{vmatrix} c \sin A & b \sin A & a^2 \\ \cos A & 1 & b \sin A \\ 1 & \cos A & c \sin A \end{vmatrix} \text{ is equal to}$$

- (A) 0 (B) $a \sin A + b \sin B + c \sin C$
(C) $(ab + bc + ca) \cos A \cos B \cos C$ (D) -1

6. If $a_1, a_2, a_3, \dots, a_n$ are positive real number forming an A.P then the value of

$$\begin{vmatrix} \ell n a_n & \ell n a_{n+2} & \ell n a_{n+4} \\ \ell n a_{n+6} & \ell n a_{n+8} & \ell n a_{n+10} \\ \ell n a_{n+12} & \ell n a_{n+14} & \ell n a_{n+16} \end{vmatrix} =$$

- (A) 1 (B) -1 (C) 0 (D) 2

7. Maximum value of a second order determinant whose each entry is either zero or one is equal to

- (A) -1 (B) 0 (C) 1 (D) 2

8. $\vec{a}_r = x_r \hat{i} + y_r \hat{j} + z_r \hat{k}, r = 1, 2, 3$ be three mutually perpendicular unit vectors then the

$$\text{value of } \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} =$$

- (A) zero (B) ± 1 (C) ± 2 (D) ± 3

9. Maximum value of the expression

$$\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 1 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix} =$$

- (A) 4 (B) 6 (C) 2 (D) -2

$$10. \Delta = \begin{vmatrix} e^x & \sin x & 1 \\ \cos x & \log_e^{(1+x^2)} & 1 \\ x & x^2 & 1 \end{vmatrix} = a + bx + cx^2$$

then $(a+b)^2$ is equal to

- A) 0 (B) 1 (C) 2 (D) 3

11. If a_i , where $i=1,2,\dots,9$ are the squares

$$\text{of odd integers then } \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} \text{ is a}$$

multiple of

- A) 9 (B) 16 (C) 64 (D) 1285

12. In a third order determinant, each element of the first column consists of sum of three terms, each element of the second column consists sum of four terms, each element of the third column consists of sum of 10 terms, then it can be decomposed into n! determinants then n=

- A) 4 (B) 5 (C) 6 (D) 120

$$13. \text{ Let } \Delta = \begin{vmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \theta \sin \phi & \sin \theta \cos \phi & 0 \end{vmatrix}$$

is independent of

- A) θ B) ϕ
C) both θ and ϕ D) its value is $\cos \theta$

14. If $\Delta_r = \begin{vmatrix} 2^{r-1} & \frac{(r+1)!}{\left(1+\frac{1}{r}\right)} & 4r^3-2nr \\ a & b & c \\ 2^n-1 & (n+1)!-1 & n^3(n+1) \end{vmatrix}$, $n \in N$ then $\sum_{r=1}^n \Delta_r =$

- A) abc B) $(n+3)!$
C) 0 D) $a(n!)+b.2^n+c$

15. If $\Delta = \begin{vmatrix} my+nz & mq-nr & mb+nc \\ kz-mx & kr-mp & kb-ma \\ nx+ky & np+kq & na+kb \end{vmatrix}$ then

- (A) $\Delta = 0$
(B) $\Delta = xyz + abc + pqr$
(C) $\Delta = f(x, y, z)$
(D) $\Delta = f(p, q, r)$

16. $\Delta_1 = \begin{vmatrix} y^5 z^6 (z^3 - y^3) & x^4 z^6 (x^3 - z^3) & x^4 y^5 (y^3 - x^3) \\ y^2 z^3 (y^6 - z^6) & xz^3 (z^6 - x^6) & xy^2 (x^6 - y^6) \\ y^2 z^3 (z^3 - y^3) & xz^3 (x^3 - z^3) & xy^2 (y^3 - x^3) \end{vmatrix}$

and $\Delta_2 = \begin{vmatrix} x & y^2 & z^3 \\ x^4 & y^5 & z^6 \\ x^7 & y^8 & z^9 \end{vmatrix}$. Then $\Delta_1 \Delta_2$ is equal to

- (A) Δ_2^3 (B) Δ_2^2 (C) Δ_2^4 (D) Δ_2^5

17. Consider the system of equation $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ and $(a_1, b_1, a_2, b_2, c_1, c_2)$ are not

all zero's) $\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$

then

Statement-I: If $\Delta = 0$ then the system is always consistent

Statement-II : If $\Delta \neq 0$ then the system is consistent

- A) Both are true B) Both are false
C) I is True, II is false
D) I is false, II is true

18. If $f(x) = ax^2 + bx + c$, $a, b, c \in R$ and equation $f(x) - x = 0$ has imaginary roots α and β and γ, δ be the roots of

$f\{f(x)\} - x = 0$ then $\begin{vmatrix} 2 & \alpha & \delta \\ \beta & 0 & \alpha \\ \gamma & \beta & 1 \end{vmatrix}$ is

- (A) 0 (B) purely real
(C) purely imaginary (D) none of these

19. If α, β, γ are real numbers then

$\Delta = \begin{vmatrix} 1 & \cos(\beta - \alpha) & \cos(\gamma - \alpha) \\ \cos(\alpha - \beta) & 1 & \cos(\gamma - \beta) \\ \cos(\alpha - \gamma) & \cos(\beta - \gamma) & 1 \end{vmatrix}$

is equal to

- (A) -1
(B) $\cos \alpha \cos \beta \cos \gamma$
(C) $\cos \alpha + \cos \beta + \cos \gamma$
(D) None of these

20. The coefficient of x in the determinant

$\begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix}$ is

- A) 2 B) 3 C) 0 D) 1

MORE THAN ONE ANSWER

21. If maximum and minimum values of

$\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix}$ are α, β then

which of the following is/are true?

- A) $\alpha + \beta^{99} = 4$
B) $\alpha^3 - \beta^{17} = 26$
C) $\alpha^{2n} - \beta^{2n}$ is always an even integer for $n \in N$
D) A triangle can be constructed having its sides as $\alpha, \beta, \alpha - \beta$

22. Let $f(x) = \begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$ then which

of the following statement(s) is/are true?

- A) period of $f(x)$ is π
 B) maximum value of $f(x)$ is 1
 C) minimum value of $f(x)$ is 0
 D) product of maximum and minimum values of $f(x)$ is 0

23. $\Delta = \begin{vmatrix} a_1 + b_1x & a_1x + b_1 & c_1 \\ a_2 + b_2x & a_2x + b_2 & c_2 \\ a_3 + b_3x & a_3x + b_3 & c_3 \end{vmatrix} = 0$ then

- (A) $x = 1$ (B) $x = -1$
 (C) $x = \pm 1$ (D) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

24. The determinant

$$\begin{vmatrix} \cos(\alpha + \beta) & \sin \alpha & -\cos \alpha \\ -\sin(\alpha + \beta) & \cos \alpha & \sin \alpha \\ \cos 2\beta & \sin \beta & \cos \beta \end{vmatrix} \text{ is}$$

- (A) non-negative (B) independent of α
 (C) independent of β
 (D) independent of both α and β

25. If $\Delta = \begin{vmatrix} 1+x & x & x^2 \\ x & 1+x & x^2 \\ x^2 & x & 1+x \end{vmatrix}$
 $= ax^5 + bx^4 + cx^3 + dx^2 + ex + f$
 (A) $f = 1$ (B) $e = 3$
 (C) $a + c = -1$ (D) $b + d = 2$

26. If

$$f(x) = \begin{vmatrix} 3 & 3x & 3x^2 + 2a^2 \\ 3x & 3x^2 + 2a^2 & 3x^3 + 4a^2x \\ 3x^2 + 2a^2 & 3x^3 + 6a^2x & 3x^4 + 8a^2x^2 + 2a^4 \end{vmatrix}$$

then

- (A) $f'(x) = 0$
 (B) $y = f(x)$ is a straight line parallel to x-axis
 (C) $\int_0^2 f(x) dx = 32a^4$
 (D) $f(x) = -16a^6$

27. If $f(x)$ and $g(x)$ are functions such that $f(x+y) = f(x)g(y) + g(x)f(y)$ then

$$\begin{vmatrix} f(\alpha) & g(\alpha) & f(\alpha + \theta) \\ f(\beta) & g(\beta) & f(\beta + \theta) \\ f(\gamma) & g(\gamma) & f(\gamma + \delta) \end{vmatrix} \text{ is}$$

independent of

- (A) α (B) β (C) γ (D) θ

28. Let $f_1(x) = x + a$, $f_2(x) = x^2 + bx + c$ and

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ f_1(x_1) & f_1(x_2) & f_1(x_3) \\ f_2(x_1) & f_2(x_2) & f_2(x_3) \end{vmatrix} \text{ then}$$

- (A) Δ is independent of a
 (B) Δ is independent of b and c
 (C) Δ is independent of x_1, x_2, x_3
 (D) none of the above

29. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$,

$$(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2,$$

$$(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2 \text{ and } 4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= \lambda \{ \lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3) \lambda^2 + (\lambda_1 \lambda_2 + \lambda_2 \lambda_3 + \lambda_3 \lambda_1) \lambda - \lambda_1 \lambda_2 \lambda_3 \}$$

then

$$(A) \lambda \geq \frac{3}{2} (\lambda_1 \lambda_2 \lambda_3)^{\frac{1}{3}} \quad (B) \lambda_1 \lambda_2 \lambda_3 = 8abc$$

$$(C) \sum \lambda_1 \lambda_2 = 4 \sum ab \quad (D) 2\lambda = \lambda_1 + \lambda_2 + \lambda_3$$

30. If

$$A = \begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha + \beta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \gamma)(\beta + \delta) & \alpha(\beta + \delta) + \beta(\alpha + \gamma) \\ \alpha + \beta & \alpha(\beta + \delta) + \beta(\alpha + \gamma) & 2\alpha\beta\delta \end{vmatrix},$$

$$B = \begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha\delta + \beta\gamma \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \delta)(\beta + \gamma) & \alpha\delta(\beta + \gamma) + \beta\gamma(\alpha + \delta) \\ \alpha\delta + \beta\gamma & \alpha\delta(\beta + \gamma) + \beta\gamma(\alpha + \delta) & 2\alpha\beta\delta \end{vmatrix}$$

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$$C = \begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha\gamma + \beta\delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \gamma)(\beta + \delta) & \alpha\gamma(\beta + \delta) \\ \alpha\gamma + \beta\delta & \alpha\gamma(\beta + \delta) + \beta\gamma(\alpha + \gamma) & 2\alpha\beta\gamma\delta \end{vmatrix}$$

then

(A) $A^3 + B^3 + C^3 = 3ABC$

(B) $A^{91} + B^{38} = C^{92} (ABC)^{54}$

(C) $\cos^A \theta + \cos^B \phi + \cos^C \psi = 3(\cos \theta)^A (\cos \phi)^B (\cos \psi)^C$

(D) none of the above

COMPREHENSION QUESTIONS

Passage - I

Consider the polynomial function

$$f(x) = \begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^a \end{vmatrix} \quad \text{and } a, b$$

being positive integers.

31. The constant term in $f(x)$ is

A) 2 B) 1 C) -1 D) 0

32. The coefficient of x in $f(x)$ is

A) 2^a B) $2^a - 3 \times 2^b = 1$ C) 0 D) 2^b

33. Which of the following is true?

A) All root of the equation $f(x) = 0$ are positive
B) All root of the equation $f(x) = 0$ are negative
C) At least one of the equation $f(x) = 0$ repeating one

D) none of these

Passage - II

Some times it is more convenient to prove a determinant identity involving symmetric and homogeneous functions by factor theorem. A dimensional check and homogeneity becomes very helpful in solving problems. Recall that an identity is true for all values of the variables. Let D_1, D_2, D_3 be determinants given by

$$D_1 = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}, D_2 = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}, D_3 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

then it can be noted that D_1, D_2, D_3 all will vanish for $a = b, b = c, c = a \Rightarrow (a - b)(b - c)(c - a)$ are factors of D_1, D_2, D_3 .

34. The remaining factor (or factors) of D_3 should be

(A) $(a^2 + b^2 + c^2)$ (B) $(ab + bc + ac)$
(C) $a^2 + b^2 - c^2$ (D) $(a - c)(b - c)$

35. The remaining factor of D_2 must be

(A) a (B) b
(C) c (D) $a + b + c$

36. If $\begin{vmatrix} -2a & a+b & c+a \\ a+b & -2b & b+c \\ c+a & b+c & -2c \end{vmatrix}$ is a product of a real

number λ and three linear factors then λ must be

(A) 2 (B) -2 (C) 4 (D) -4

MATRIX MATCH QUESTIONS

37. Observe the following columns
Column-I

(A) If $a^2 + b^2 + c^2 = 1$ and

$$\Delta = \begin{vmatrix} a^2 + (b^2 + c^2)\cos\phi & ab(1 - \cos\phi) & ac(1 - \cos\phi) \\ ba(1 - \cos\phi) & b^2 + (c^2 + a^2)\cos\phi & bc(1 - \cos\phi) \\ ca(1 - \cos\phi) & ab(1 - \cos\phi) & c^2 + (a^2 + b^2)\cos\phi \end{vmatrix}$$

then Δ is independent of

(B) $\Delta = \begin{vmatrix} \sin a & \cos a & \sin(a + \phi) \\ \sin b & \cos b & \sin(b + \phi) \\ \sin c & \cos c & \sin(c + \phi) \end{vmatrix}$

then Δ is independent of

(C) If

$$\Delta = \begin{vmatrix} 1/c & 1/c & -(a+b)^2/c^2 \\ -(b+c)/a^2 & 1/a & 1/a \\ -\frac{b(b+c)}{a^2c} & \frac{(a+2b+c)}{ac} & -\frac{(a+b)b}{ac^2} \end{vmatrix}$$

then Δ is independent of

Column-II

(P) a
(Q) b
(R) c
(S) ϕ
(T) $\Delta = 0$

38. Match the following:

Column-I

A) Coefficient of x in

$$f(x) = \begin{vmatrix} x & (1 + \sin x)^3 & \cos x \\ 1 & \log(1+x) & 2 \\ x^2 & 1+x^2 & 0 \end{vmatrix} \text{ is}$$

B) Maximum value of

$$\begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix} \text{ is}$$

C) If a, b, c are in A.P. and

$$f(x) = \begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$$

then $f'(0)$ is

$$\text{D) If } r > 1, M_r = \begin{vmatrix} r-1 & \frac{1}{r} \\ 1 & \frac{1}{(r-1)^2} \end{vmatrix} \text{ (s) -2}$$

39. Match the following

Column-I

(A) The value of the

$$\begin{vmatrix} x+2 & x+3 & x+5 \\ x+4 & x+6 & x+9 \\ x+8 & x+11 & x+15 \end{vmatrix} \text{ is}$$

(B) If one of the roots of the equation

$$\begin{vmatrix} 7 & 6 & x^2-13 \\ 2 & x^2-13 & 2 \\ x^2-13 & 3 & 7 \end{vmatrix} = 0$$

is -2, then sum of all other five roots is

(C) The value of

$$\begin{vmatrix} \sqrt{6} & 2i & 3+\sqrt{6} \\ \sqrt{12} & \sqrt{3}+\sqrt{8}i & 3\sqrt{2}+\sqrt{6}i \\ \sqrt{18} & \sqrt{2}+\sqrt{12}i & \sqrt{27}+2i \end{vmatrix} \text{ is}$$

(D) If

$$f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$$

then the value of $f(\pi/3)$ is

Column-II

(p) 10

(q) 0

(r) -12

Column-II

(p) 1

(q) -6

(r) 2

(s) -2

40. Let a, b, c be positive integers such that

$$\begin{vmatrix} b+c & c & b \\ c & c+a & a \\ b & a & a+b \end{vmatrix} = 4 \text{ then}$$

List-I

a) $(a+b-c)^3 =$

b) $a^3+b^3+c^3-3abc =$

c) $\frac{1}{4}(a+b)(b+c)(c+a) =$

d) $a^3b^2c+ab^2c^3+a^2bc^3 =$

List-II

p) 0

q) 1

r) 2

s) 3

41. List-I

$$\text{a) If } r > 1, M_r = \begin{vmatrix} r-1 & \frac{1}{r} \\ 1 & \frac{1}{(r-1)^2} \end{vmatrix} \text{ then p) 2}$$

$$\lim_{n \rightarrow \infty} (|M_2| + |M_3| + |M_4| + \dots + |M_n|)^{\log n} =$$

$$\text{b) If } \begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix} \text{ q) 4}$$

then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma =$

$$\text{c) If } A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}, C = (BAB^{-1})(B^{-1}A^TB) \text{ r) 1}$$

then $\sqrt{\det(C)} =$

$$\text{d) If } A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \text{ and s) 3}$$

$$A^4 = -\lambda I \text{ then } \lambda =$$

42. List-I

A) If

$$\begin{vmatrix} x & x+y & x+y+z \\ 2x & 3x+2y & 4x+3y+2z \\ 3x & 6x+3y & 10x+6y+3z \end{vmatrix} = 343$$

then $x =$

$$\text{B) If } a^2+b^2+c^2 = -2 \text{ and q) 4}$$

$$f(x) = \begin{vmatrix} (1+a^2x) & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & (1+b^2x) & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & (1+c^2x) \end{vmatrix}$$

then $f(x)$ is a polynomial of degree

$$\text{C) } \begin{vmatrix} \frac{a^2+b^2}{c} & c & c \\ a & \frac{b^2+c^2}{a} & a \\ b & b & \frac{c^2+a^2}{b} \end{vmatrix} \text{ If}$$

$$= kabc \text{ then } k =$$

r) 0

DETERMINANTS

D) If A, B, C are the angles of Δ then

$$\text{then } \begin{vmatrix} \sin 2A & \sin C & \sin B \\ \sin C & \sin 2B & \sin A \\ \sin B & \sin A & \sin 2C \end{vmatrix} =$$

s) 7

ASSERTION & REASON QUESTIONS

(A) If both (A) and (R) are true and (R) is correct explanation of (A).

(B) If both (A) and (R) are true but (R) is not correct explanation of (A).

(C) If (A) is true, (R) is false

(D) If (A) is false, (R) is true

43. (A) If $D(x) = \begin{vmatrix} f_1(x) & f_2(x) & f_3(x) \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ where

f_1, f_2, f_3 are differentiable functions and $a_2, b_2, c_2, a_3, b_3, c_3$ are constants then

$$\int D(x) dx = \begin{vmatrix} \int f_1(x) & \int f_2(x) & \int f_3(x) \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + c$$

(R) Integration of sum of several functions is equal to sum of integration of individual functions.

44. (A) If A is a skew symmetric of order 3 then its determinants should be zero.

(R) If A is square matrix then

$$\det A = \det A' = \det(-A')$$

45. (A) If three digit numbers A28, 3B9 and 62C, where A, B and C are integers between

0 and 9, be divisible by a fixed integer k then

$$\begin{vmatrix} A & 3 & 6 \\ 8 & 9 & C \\ 2 & B & 2 \end{vmatrix} \text{ is divisible by } k.$$

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(R) If sum of three numbers is divisible by k then atleast one of them must be divisible by k.

46. (A) For all θ

$$\begin{vmatrix} \sin \theta & \cos \theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix} = 0$$

$$(R) \sin\left(\theta + \frac{2\pi}{3}\right) + \sin\left(\theta - \frac{2\pi}{3}\right) = -\sin \theta$$

$$\cos\left(\theta + \frac{2\pi}{3}\right) + \cos\left(\theta - \frac{2\pi}{3}\right) = -\cos \theta$$

$$\sin\left(2\theta + \frac{4\pi}{3}\right) + \sin\left(2\theta - \frac{4\pi}{3}\right) = -\sin 2\theta$$

47. (A) If a, b, c are different and

$$\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = 0 \text{ then}$$

$$abc(ab + bc + ac) = a + b + c$$

$$(R) \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (ab + bc + ac) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

INTEGER QUESTIONS

48. The set of equations $2x - 3y = 4$, $7x - 2y = 2$, $9x - 5y = [4a]$ may be consistent, then [a] is (where $[\cdot]$ denotes G.I.F)

49. If $f(x)$ satisfies the equation

$$\begin{vmatrix} f(x-2) & f(x+5) & f(x-2) \\ 5 & 4 & -5 \\ 10 & 12 & 30 \end{vmatrix} = 0 \text{ for all}$$

real x then $f(x)$ is periodic with period

50. If A is a square matrix of order n, such that $|\text{adj}(\text{adj} A)| = |A|^9$ then n is

$$51. \text{ If } \Delta = \begin{vmatrix} e^x & \sin x & 1 \\ \cos x & \ln(1+x^2) & 1 \\ x & x^2 & 1 \end{vmatrix} = a + bx + cx^2$$

then the value of a-b is

52. If $D = \begin{vmatrix} 10! & 11! & 12! \\ 11! & 12! & 13! \\ 12! & 13! & 14! \end{vmatrix}$ then $k/3$, where k is

the total number of divisor of $\frac{D}{(10!)^3} - 4$, is

KEY

LEVEL -V

SINGLE ANSWER QUESTIONS

- 1.B 2.B 3.B 4.C
5.A 6.C 7.C 8.B
9.B 10.B 11.C 12.B
13.B 14.C 15.A 16.A
17.D 18.D 19.D 20.C

MORE THAN ONE ANSWER

- 21.ABC 22.ABC 23.ABCD 24.AB
25.ABC 26.AB 27.ABCD 28.AB
29.ABCD 30.ABC

COMPREHENSION QUESTIONS

- 31.D 32.C 33.C 34.B
35.D 36.C

MATRIX MATCH QUESTIONS

37. $A \rightarrow P, Q, R; B \rightarrow P, Q, R, S, T; C \rightarrow P, Q, R, S, T$
38. A-S, B-P, C-Q, D-S
39. A-S, B-R, C-Q, D-P
40. A-Q, B-P, C-R, D-S
41. A-R, B-P, C-Q, D-Q
42. A-S, B-P, C-Q, D-R

ASSERTION REASON QUESTIONS

43. A 44. C 45. C 46. A
47. A

INTEGERS QUESTIONS

- 48.1 49.7 50.4 51. 1
52. 6

HINTS

LEVEL -V

SINGLE ANSWER QUESTIONS

1. $\frac{d}{dx} \Delta_1 = \begin{vmatrix} 1 & 0 & 0 \\ a & x & b \\ a & a & x \end{vmatrix} + \begin{vmatrix} x & b & b \\ 0 & 1 & 0 \\ a & a & x \end{vmatrix} + \begin{vmatrix} x & b & b \\ a & x & b \\ 0 & 0 & 1 \end{vmatrix}$

$$= \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix} + \begin{vmatrix} x & b \\ a & x \end{vmatrix} = 3\Delta_2$$

2. $C_3 \rightarrow C_3 - xC_1 - yC_2$
 $\Rightarrow -(ax^2 + 2bxy + cy^2)(ac - b^2)$
 $-\frac{1}{a}(a^2x^2 + 2baxy + acy^2 + b^2y^2 - b^2y^2)(ac - b^2)$
 $-\frac{1}{a}\{(ax + by)^2 + y^2(ac - b^2)\}(ac - b^2) < 0$

3. $= e^{-i2A}(e^{-i2(B+C)} - e^{i2A}) - e^{iC}(e^{-iC} - e^{i(A+B)}) + e^{iB}(e^{i(A+C)} - e^{iB})$
 $= e^{-i2(A+B+C)} - 1 - 1 + e^{i(A+B+C)} + e^{i(A+B+C)} - 1$
 $= e^{-i2\pi} - 2 + e^{i\pi} + e^{i\pi} - 1 = -4$

4. $\Delta = \begin{vmatrix} 0 & 0 & 1 \\ 1-a & b-1 & 1 \\ 1-ac & 1-c & c \end{vmatrix}$
 $[C_1 \rightarrow C_1 - aC_3; C_2 \rightarrow C_2 - C_3]$
 $= (1-a)(1-c) - (b-1)(1-ac)$
 $= 1 - a - c + ac - b + abc + 1 - ac$
 $= (2 + abc) - (a + b + c)$

According to the given condition, we have $D > 0$

i.e. $abc + 2 > a + b + c$

using AM > G.M., we have

$$a + b + c > 3(abc)^{1/3}$$

Thus, we have $abc + 2 > 3(abc)^{1/3}$

$$\Rightarrow x^3 - 3x + 2 > 0$$

$$(x-1)^2(x+2) > 0, \text{ i.e. } x > -2$$

$$\text{i.e. } x^3 = abc > -8$$

5. $D = c \sin A (c \sin A - b \sin A \cos A)$
 $- b \sin A (c \sin A \cos A - b \sin A) + a^2 (\cos^2 A - 1)$
 $= (c^2 + b^2 - a^2) \sin^2 A - 2bc \sin^2 A \cos A$
 $= (2bc \cos A) \sin^2 A - 2bc \sin^2 A \cos A$
 $= 0 \quad [a^2 = b^2 + c^2 - 2bc \cos A]$

6. Let r be the common ratio of G.P.

$$a_i = ar^{i-1} \forall i \geq 1$$

$$\ell na_i = \ell na + (i-1)\ell nr$$

$$= \alpha + (i-1)\beta \quad \text{where } \alpha = \ell na \text{ and}$$

$$\beta = \ell nr$$

$$= \begin{vmatrix} \alpha + (n-1)\beta & \alpha + (n+1)\beta & \alpha + (n-3)\beta \\ \alpha + (n+5)\beta & \alpha + (n+7)\beta & \alpha + (n+9)\beta \\ \alpha + (n+11)\beta & \alpha + (n+13)\beta & \alpha + (n+15)\beta \end{vmatrix}$$

Applying $C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_2$,

$$\Delta = \begin{vmatrix} \alpha + (n+1)\beta & 2\beta & 2\beta \\ \alpha + (n+5)\beta & 2\beta & 2\beta \\ \alpha + (n+11)\beta & 2\beta & 2\beta \end{vmatrix} = 0$$

DETERMINANTS

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$$7. \Delta = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = (a_1 b_2 - a_2 b_1)$$

$a_1 b_2 = 0$ or 1 and $a_2 b_1 = 0$ or 1
Max. $(\Delta) = 1$

$$8. \Delta = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$\Delta^2 = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$$

$$= \begin{vmatrix} x_1^2 + y_1^2 + z_1^2 & x_1 x_2 + y_1 y_2 + z_1 z_2 & x_1 x_3 + y_1 y_3 + z_1 z_3 \\ x_1 x_2 + y_1 y_2 + z_1 z_2 & x_2^2 + y_2^2 + z_2^2 & x_2 x_3 + y_2 y_3 + z_2 z_3 \\ x_1 x_3 + y_1 y_3 + z_1 z_3 & x_2 x_3 + y_2 y_3 + z_2 z_3 & x_3^2 + y_3^2 + z_3^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

$$9. \Delta^2 = 1 \Rightarrow \Delta = \pm 1$$

Applying $C_1 \rightarrow C_1 + C_2$

$$\Delta = \begin{vmatrix} 2 & \cos^2 x & 4 \sin 2x \\ 2 & 1 + \cos^2 x & 4 \sin 2x \\ 1 & \cos^2 x & 1 + 4 \sin 2x \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$

$$\Delta = \begin{vmatrix} 2 & \cos^2 x & 4 \sin 2x \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 2 + 4 \sin 2x \leq 6$$

$$10. \text{ Put } x=0 \Rightarrow a=0, \text{ differentiate and } x=0 \Rightarrow b=1$$

$\therefore (a+b)^2 = 1$

$$11. \begin{vmatrix} (2m+1)^2 & (2n+1)^2 & (2p+1)^2 \\ (2q+1)^2 & (2r+1)^2 & (2s+1)^2 \\ (2t+1)^2 & (2u+1)^2 & (2v+1)^2 \end{vmatrix}, \Rightarrow C_1 \rightarrow C_1 - C_2,$$

$$C_2 \rightarrow C_2 - C_3$$

$$C_1 - C_2 \Rightarrow (2m+1+2n+1)(2m+1-2n-1) \Rightarrow 2.2(\quad) = 4(\quad)$$

Similarly $C_2 - C_3 = 4(\quad)$

$$\Rightarrow 4.4 \begin{vmatrix} (m+n+2)(m-n) & (n+p+2)(n-p) & (2p+1)^2 \\ (q+r+2)(q-r) & (r+s+2)(r-s) & (2s+1)^2 \\ (t+u+2)(t-u) & (u+v+2)(u-v) & (2v+1)^2 \end{vmatrix}$$

$$\Rightarrow R_1 \rightarrow R_1 - R_2$$

$$12. n! = 3.4.10 = 120 = 5!$$

$$13. C_1 \rightarrow C_1 - \frac{\cos \phi}{\sin \phi} \cdot C_2$$

$$\Rightarrow \Delta = \frac{-\sin \theta}{\sin \phi} (-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi) = \sin \theta$$

$$14. R_1 \text{ \& } R_3 \text{ are identical after applying } \Sigma$$

$$15. \Delta \text{ can be written as}$$

$$\Delta = \begin{vmatrix} my + nz & kz - mx & nx + ky \\ mq - nr & kr - mp & np + kq \\ mb + nc & kb - ma & na + kb \end{vmatrix}$$

$$= - \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} \times \begin{vmatrix} 0 & m & n \\ -m & 0 & k \\ -n & -k & 0 \end{vmatrix} = 0$$

$$\left(\begin{vmatrix} 0 & m & n \\ -m & 0 & k \\ -n & -k & 0 \end{vmatrix} \text{ is skew symmetric} \right)$$

$$16. \text{ The given determinants } \Delta_1 \text{ is obtained by corresponding cofactors of determinant } \Delta_2;$$

Hence $\Delta_1 = \Delta_2^2$.

Now $\Delta_1 \Delta_2 = \Delta_2^2$, $\Delta_2 = \Delta_2^3$.

$$17. \text{ Statement I is sometimes inconsistent } (\because \text{no solution}) \text{ and Statement II is always true}$$

$$18. \because f(x) - x = 0 \text{ has imaginary roots,}$$

then $f(x) - x > 0$ or $f(x) - x < 0, \forall x \in R$

for $f(x) - x > 0, \forall x \in R$

then $f\{f(x)\} - f(x) > 0, \forall x \in R$

adding we get, $f\{f(x)\} - x > 0, \forall x \in R$

Similarly, $f\{f(x)\} - x < 0, \forall x \in R$

Thus, roots of equation $f\{f(x)\} - x = 0$ are imaginary.

$$z = \begin{vmatrix} 2 & \alpha & \delta \\ \beta & 0 & \alpha \\ \gamma & \beta & 1 \end{vmatrix}$$

$$\text{Then } \bar{z} = \begin{vmatrix} 2 & \bar{\alpha} & \bar{\delta} \\ \bar{\beta} & 0 & \alpha \\ \bar{\gamma} & \bar{\beta} & 1 \end{vmatrix} = \begin{vmatrix} 2 & \beta & \gamma \\ \alpha & 0 & \beta \\ \delta & \alpha & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & \alpha & \delta \\ \beta & 0 & \alpha \\ \gamma & \beta & 1 \end{vmatrix} = z$$

Hence, z is purely real.

$$19. \Delta = \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix} \times \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix}$$

$$= 0 \times 0 = 0$$

$$20. \begin{vmatrix} (1+x)^{a_1b_1} & (1+x)^{a_1b_2} & (1+x)^{a_1b_3} \\ (1+x)^{a_2b_1} & (1+x)^{a_2b_2} & (1+x)^{a_2b_3} \\ (1+x)^{a_3b_1} & (1+x)^{a_3b_2} & (1+x)^{a_3b_3} \end{vmatrix} = A + Bx + Cx^2 + \dots$$

Differentiating both sides w.r.t. x and then put $x = 0$, we get

$$\begin{vmatrix} a_1b_1 & a_2b_2 & a_3b_3 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ a_2b_1 & a_2b_2 & a_2b_3 \\ 1 & 1 & 1 \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ a_3b_1 & a_3b_2 & a_3b_3 \end{vmatrix} = B$$

$$\Rightarrow 0 + 0 + 0 = B$$

$$\therefore B = 0$$

Hence, coefficient of $x=0$

MORE THAN ONE ANSWER

$$21. f(x) = 2 + \sin 2x \text{ (by simplifying)}$$

& $\alpha = 3$ (maximum value of $f(x)$)

$\beta = 1$ (minimum value of $f(x)$)

$$22. \text{ by simplifying}$$

$$f(x) = \cos^2 x - \cos^2 x \sin^2 x = \cos^4 x$$

$$\therefore \text{ period of } f(x) = \pi$$

Maximum of $f(x) = 1$ and minimum of $f(x) = 0$

$$23. \text{ Check by option for } x = 1 \text{ and } x = -1.$$

$$24. \begin{vmatrix} \cos(\alpha + \beta) & \sin \alpha & -\cos \alpha \\ -\sin(\alpha + \beta) & \cos \alpha & \sin \alpha \\ \cos 2\beta & \sin \beta & \cos \beta \end{vmatrix}$$

$$= (\cos(\alpha + \beta)(\cos \alpha \cos \beta - \sin \alpha \sin \beta) + \sin \alpha (\sin \alpha \cos 2\beta + \cos \beta \sin(\alpha + \beta)) - \cos \alpha (-\sin \beta \sin(\alpha + \beta)))$$

$$- \cos \alpha \cos 2\beta$$

$$= \cos^2(\alpha + \beta) + \cos 2\beta + \sin^2(\alpha + \beta)$$

$$= 1 + \cos 2\beta$$

Hence the value of the given determinant is non-negative and independent of α .

$$25. \text{ Put } x = 0 \Rightarrow f = \Delta(0) = 1$$

$$e = \Delta'(0) = 3$$

$$\text{Put } x = 1$$

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = a + b + c + d + e + f$$

$$\therefore a + b + c + d = 0 \dots(i)$$

$$\text{Put } x = -1$$

$$\Rightarrow -a + b - c + d - 3 + 1 = 0$$

$$-a + b - c + d = 2 \dots(ii)$$

On solving equation (i) & (ii),

we get $b + d = 1$ and $a + c = -1$

$$26. \text{ Applying } C_3 \rightarrow C_3 - xC_2, C_2 \rightarrow C_2 - xC_1, \text{ we obtain}$$

$$\Delta(x) = \begin{vmatrix} 3 & 0 & 2a^2 \\ 3x & 2a^2 & 2a^2x \\ 3x^2 + 2a^2 & 4a^2x & 2a^2x^2 + 4a^4 \end{vmatrix}$$

$$\text{Applying } C_3 \rightarrow C_3 - xC_2, \text{ we get}$$

$$\Delta(x) = 4a^4 \begin{vmatrix} 3 & 0 & 1 \\ 3x & 1 & x \\ 3x^2 + 2a^2 & 2x & x^2 + 2a^2 \end{vmatrix}$$

$$\text{Applying } C_1 \rightarrow C_1 - 3C_3, \text{ we get}$$

$$\Delta(x) = 4a^4 \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & x \\ -4a^2 & 2x & x^2 + 2a^2 \end{vmatrix} = 16a^6$$

$$27. \text{ Now, applying } C_3 \rightarrow C_3 - g(\theta)C_1 - f(\theta)C_2$$

$$\begin{vmatrix} f(\alpha) & g(\alpha) & 0 \\ f(\beta) & g(\beta) & 0 \\ f(\gamma) & g(\gamma) & 0 \end{vmatrix} = 0$$

$$[\because f(x+y) = f(x)g(y) + g(x)f(y)]$$

$$28. \text{ Applying } C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1, \text{ then}$$

DETERMINANTS

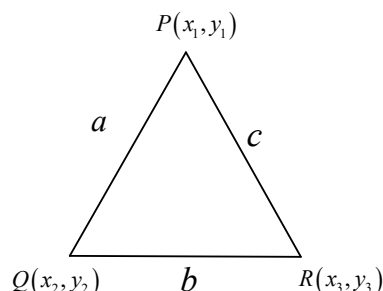
$$\Delta = \begin{vmatrix} 1 & \dots & 0 & \dots & 0 \\ \vdots & & & & \\ f_1(x_1) & f_1(x_2) - f_2(x_1) & f_1(x_3) - f_2(x_1) \\ \vdots & & \\ x_2 - x_1 & & x_3 - x_1 \end{vmatrix}$$

$$= \begin{vmatrix} x_2 - x_1 & x_3 - x_1 \\ (x_2 - x_1)(x_1 + x_2 + b) & (x_3 - x_1)(x_1 + x_3 + b) \end{vmatrix}$$

$$= (x_2 - x_1)(x_3 - x_1) \begin{vmatrix} 1 & 1 \\ x_1 + x_2 + b & x_1 + x_3 + b \end{vmatrix}$$

$$= (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$$

29. Area of $\Delta PQR = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$



$$= \sqrt{\frac{(a+b+c)}{2} \cdot \frac{(a+b-c)}{2} \cdot \frac{(b+c-a)}{2} \cdot \frac{(c+a-b)}{2}}$$

$$4 \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = (a+b+c)(b+c-a)(c+a-b)(a+b-c)$$

i.e. $a+b+c = \lambda, 2a = \lambda_1, 2b = \lambda_2, 2c = \lambda_3$

$\therefore 2\lambda = \lambda_1 + \lambda_2 + \lambda_3$

$\sum \lambda_1 \lambda_2 = 4(ab + bc + ca) = 4 \sum ab$

$\lambda_1 \lambda_2 \lambda_3 = 8abc$ and $\frac{a+b+c}{3} \geq (abc)^{1/3}$

$\Rightarrow \frac{\lambda}{3} \geq \left(\frac{\lambda_1}{2} \cdot \frac{\lambda_2}{2} \cdot \frac{\lambda_3}{2} \right)^{1/3} \Rightarrow \lambda \geq \frac{3}{2} (\lambda_1 \lambda_2 \lambda_3)^{1/3}$

30. $A = \begin{vmatrix} 1 & 1 & 0 \\ \gamma + \delta & \alpha + \beta & 0 \\ \gamma \delta & \alpha \beta & 0 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 0 \\ \alpha + \beta & \gamma + \delta & 0 \\ \alpha + \beta & \gamma \delta & 0 \end{vmatrix} = 0 \times 0 = 0$

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Similarly, $B = 0, C = 0$

$\therefore A + B + C = 0 \Rightarrow A^3 + B^3 + C^3 = 3ABC$

COMPREHENSION QUESTIONS

31,32,33

Let,

$$\begin{vmatrix} (1+x)^a & (1+2x)^b & 1 \\ 1 & (1+x)^a & (1+2x)^b \\ (1+2x)^b & 1 & (1+x)^a \end{vmatrix} = A + Bx + Cx^2 + \dots$$

Putting $x=0$, we get

$$A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Now differentiating both sides with respect to x and putting $x=0$, we get

$$B = \begin{vmatrix} a & 2b & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 0 & a & 2b \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 2b & 0 & a \end{vmatrix} = 0$$

34. $(a-b)(b-c)(c-a)$ does not change if a is replaced by b , b by c and c by a . But the degree of expression is 5, therefore it must have a homogeneous symmetric expression of degree 2 as factor. The two degree symmetric expressions are only

$a^2 + b^2 + c^2, ab + bc + ca, (a+b+c)^2$

$a^2 + b^2 + c^2 - ab - bc - ac$.

Thus one of the following must be an identity.

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = k(a-b)(b-c)(c-a)(a^2 + b^2 + c^2) \dots (i)$$

$$\Rightarrow \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = k(a-b)(b-c)(c-a)(a+b+c)^2 \dots (ii)$$

$$\Rightarrow \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = k(a-b)(b-c)(c-a)(a^2 + b^2 + c^2 - ab - bc - ac) \dots (iii)$$

$$\Rightarrow \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = k(a-b)(b-c)(c-a)(ab+bc+ac) \dots (iv)$$

Where k is a constant.

In the first assumption putting $a = 0, b = 1, c = -1$, we get

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = k(0-1)(1+1)(-1-0)(2)$$

$$\Rightarrow -2 = 4k \Rightarrow k = -\frac{1}{2}$$

Also putting $a = 0, b = 1, c = 2$, we get $k = \frac{5}{2}$

Since the two values of k are different;

(i) cannot be an identity.

We can similarly show that second and third are not identities either. Thus the only possibility left is fourth.

Thus

$$\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = k(a-b)(b-c)(c-a)(ab+bc+ac)$$

On putting $a = 0, b = 1, c = -1$, we get $k = 1$ (any other choice of values for a, b, c will yield same k).

35. Since the degree of D_2 is 4, the fourth factor should be of degree 1. This factor should be symmetric i.e., if $a \rightarrow b, b \rightarrow c$ and $c \rightarrow a$, the value of the factor must not change (since in D_2 if $a \rightarrow b, b \rightarrow c, c \rightarrow a$, then no change occurs.)

The only 1 dimensional factor which is symmetric in a, b, c is $a + b + c$.

36. On putting $a = -b$, the determinant

$$\begin{aligned} &= \begin{vmatrix} 2b & 0 & c-b \\ 0 & 2b & b+c \\ c-b & b+c & -2c \end{vmatrix} \\ &= 2b[4bc - (b+c)^2] + (c-b)[2b(c-b)] \\ &= 2b(b-c)^2 - 2b(b-c)^2 = 0 \end{aligned}$$

$\Rightarrow a + b$ is a factor. Since the determinant is symmetric, $b + c, c + a$ should also be the factors since the determinant is of degree 3. No other factor is possible.

$$\Rightarrow \begin{vmatrix} -2a & a+b & c+a \\ a+b & -2b & b+c \\ c+a & b+c & 2c \end{vmatrix} = k(a+b)(b+c)(c+a)$$

Where k is a constant.

Since this is an identity, we put $a = b = c = 1$

To get k , we get

$$\begin{vmatrix} -2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{vmatrix} = 8k \Rightarrow 32 = 8k \Rightarrow k = 4$$

MATRIX MATCH QUESTIONS

37. Conceptual
38. (A) Coefficient of x in $f(x)$ is coefficient of x in

$$\begin{vmatrix} x & 1 & 1 \\ 1 & x & 2 \\ x^2 & 1 & 0 \end{vmatrix}$$

Therefore, coefficient of x is -2 .

$$\begin{aligned} \text{(B) Let } D &= \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix} \\ &= (3\cos\theta - \sin\theta)^2 \\ D_{\max} &= 10 \end{aligned}$$

$$\text{(C) } f'(x) = 0$$

$$\Rightarrow f'(0) = 0$$

$$\text{(D) } a_0 = \begin{vmatrix} 0 & 2 & 0 \\ 1 & 0 & 6 \\ 0 & 0 & 1 \end{vmatrix} = -2(1) = -2$$

39. (A) The given determinant is

$$\Delta = \begin{vmatrix} x+2 & x+3 & x+5 \\ x+4 & x+6 & x+9 \\ x+8 & x+11 & x+15 \end{vmatrix}$$

Applying $R_3 \rightarrow R_3 - R_2$ and $R_2 \rightarrow R_2 - R_1$, we have

$$\Delta = \begin{vmatrix} x+2 & x+3 & x+5 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{vmatrix} = 2 \begin{vmatrix} x & x & x+1 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{vmatrix}$$

[Applying $R_1 \rightarrow R_1 - R_2$ and $R_3 \rightarrow R_3 - R_2$]

$$= 2 \begin{vmatrix} x & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

[Applying $C_3 \rightarrow C_3 - C_2$ and $C_2 \rightarrow C_2 - C_1$] = -2
[Expanding along R_3]

$$(B) \begin{vmatrix} 7 & 6 & x^2 - 13 \\ 2 & x^2 - 13 & 2 \\ x^2 - 13 & 3 & 7 \end{vmatrix}$$

Let $x^2 - 13 = t$. Then $t^3 - 67t + 126 = 0$

$$\Rightarrow t = -9, 2, 7 \Rightarrow x = \pm 2, \pm \sqrt{20}, \pm \sqrt{15}$$

Hence sum of other five roots is 2.

$$(C) \Delta \begin{vmatrix} \sqrt{6} & 2i & 3 + \sqrt{6} \\ \sqrt{12} & \sqrt{3} + \sqrt{8}i & 3\sqrt{2} + \sqrt{6}i \\ \sqrt{18} & \sqrt{2} + \sqrt{12}i & \sqrt{27} + 2i \end{vmatrix}$$

Taking $\sqrt{6}$ common from C_1 , we get

$$\Delta = \sqrt{6} \begin{vmatrix} 1 & 2i & 3 + \sqrt{6} \\ \sqrt{2} & \sqrt{3} + 2\sqrt{2}i & 3\sqrt{2} + \sqrt{6}i \\ \sqrt{3} & \sqrt{2} + 2\sqrt{3}i & 3\sqrt{3} + 2i \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - \sqrt{2} R_1$ and $R_3 \rightarrow R_3 - \sqrt{3} R_1$, we get

$$\Delta = \sqrt{6} \begin{vmatrix} 1 & 2i & 3 + \sqrt{6} \\ 0 & \sqrt{3} & \sqrt{6}i - 2\sqrt{3} \\ 0 & \sqrt{2} & 2i - 3\sqrt{2} \end{vmatrix}$$

$$= \sqrt{6}(-3\sqrt{6} + 2\sqrt{6})$$

= -6, which is an integer

$$(D) f(\theta) = \begin{vmatrix} \cos^2 \theta & \cos \theta \sin \theta & -\sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$$

Applying $R_1 \rightarrow R_1 + (\sin \theta) R_3$ and

$R_2 \rightarrow R_2 - (\cos \theta) R_3$, we get

$$f(\theta) = \begin{vmatrix} 1 & 0 & -\sin \theta \\ 0 & 1 & \cos \theta \\ \sin \theta & -\cos \theta & 0 \end{vmatrix}$$

$$= \sin^2 \theta + \cos^2 \theta = 1$$

$$\Rightarrow f(\pi/3) = 1$$

$$40. \text{ Given determinant } \Rightarrow 4abc = 4$$

$$\Rightarrow a = 1, b = 1, c = 1$$

($\because a, b, c$ are positive integers)

$$41. a) |M_r| = \frac{1}{r-1} - \frac{1}{r}$$

$$\therefore |M_2| + |M_3| + \dots + |M_n| = 1 - \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} (|M_2| + |M_3| + \dots + |M_n|) = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^{\log_e n} = 1$$

b) 2

$$c) |C| = |A|^2 = 16$$

$$d) A^2 = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} \text{ and } A^4 = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix}$$

$$42. a) x^3 = 343 \Rightarrow x = 7$$

$$b) C_1 \rightarrow C_1 + C_2 + C_3 \text{ and } \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} \text{ then}$$

$$\Delta = (1-x)^2$$

$$c) \Delta = 4abc$$

d) Verify for equilateral triangle then the result is 0

ASSERTION REASON QUESTIONS

$$43. D(x) = \lambda_1 f_1(x) - \lambda_2 f_2(x) + \lambda_3 f_3(x)$$

$$\text{Where } \lambda_1 = (b_2 c_3 - b_3 c_2), \lambda_2 = a_2 c_3 - a_3 c_2$$

$$\lambda_3 = a_2 b_3 - a_3 b_2 \text{ then}$$

$$\int D(x) dx = \int \lambda_1 f_1(x) dx + \int \lambda_2 f_2(x) dy + \int \lambda_3 f_3(x) + C. (*)$$

$$= \begin{vmatrix} \int f_1(x) dx & \int f_2(x) dx & \int f_3(x) dx \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + C$$

Thus assertion (A) is true and follows from reason R which we have applied at (*).

44. The reason (R) is false since $\det A' = \det(-A')$ is not true

$$\text{Indeed } \det(-A') = (-1)^3 \det A'$$

Now as $A = -A'$ (A is skew symmetric $\det A = \det(-A') = -\det(A') = -\det A$)

$$\Rightarrow \det A = 0$$

$$45. \text{ Operate } R_2 \rightarrow R_2 + 10R_3 + 100R_3$$

$$D = \begin{vmatrix} A & 3 & 6 \\ A28 & 3B9 & 62C \\ 2 & B & 2 \end{vmatrix}$$

As A28, 3B9, 62C are divisible by a fixed integer K, therefore

$$A28 = Ka, 3B9 = Kb, 62C = Kg$$

$$D = K \begin{vmatrix} A & 3 & 6 \\ \alpha & \beta & \gamma \\ 2 & B & 2 \end{vmatrix}$$

$\Rightarrow D$ is divisible by K.

Reason (R) is false. Take three number 3, 8, 10.

Sum is 21 which is divisible by 7 but none of them is divisible by 7.

46. Operate $R_2 \rightarrow R_2 + R_3$ and used C-D formulae as given in Reason (R) we can get result.

$$47. D = \begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} + \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix}$$

$$D = abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^3 & c^3 \end{vmatrix} + \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = 0$$

$$D = abc(D_1) + (D_2) = 0 \quad \dots(i)$$

Now

$$D_1 = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^3 & c^3 \end{vmatrix} = abc \begin{vmatrix} \frac{1}{a} & a & a^2 \\ \frac{1}{b} & b & b^2 \\ \frac{1}{c} & c & c^2 \end{vmatrix} = \begin{vmatrix} bc & a & a^2 \\ ca & b & b^2 \\ ab & c & c^2 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_3 = \begin{vmatrix} bc - a^2 & a & a^2 \\ ca - b^2 & b & b^2 \\ ab - c^2 & c & c^2 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_3 = \begin{vmatrix} bc + ac - a^2 & a & a^2 \\ bc + ac - b^2 & b & b^2 \\ ab & c & c^2 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - aC_2$$

$$D_1 = \begin{vmatrix} bc + ac & a & a^2 \\ bc + ac + ab - b^2 & b & b^2 \\ ab + bc & c & c^2 \end{vmatrix}$$

$$C_1 \rightarrow C_1 + bC_2 = \begin{vmatrix} ab + bc + ca & a & a^2 \\ ab + bc + ca & b & b^2 \\ ab + bc + ca & c & c^2 \end{vmatrix} = (ab + bc + ca) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

After operating $R_2 \rightarrow R_2 - R_3, R_3 \rightarrow R_3 - R_1$
We get $= (ab + bc + ca)(a - b)(b - c)(c - a) \dots(ii)$

$$D_2 = \begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix}$$

After operating $R_2 \rightarrow R_2 - R_3, R_3 \rightarrow R_3 - R_1$
We get $D_2 = (a - b)(b - c)(c - a)(a + b + c) \dots(iii)$

From (i), (ii) and (iii), we get

$$abc(ab + bc + ca) = a + b + c$$

($\because a, b, c$ are different)

48.

$$\begin{vmatrix} 2 & -3 & -4 \\ 7 & -2 & -2 \\ 9 & -5 & -[4a] \end{vmatrix} = 0$$

Applying $C_3 \rightarrow C_3 - C_2$, then

$$\begin{vmatrix} 2 & -3 & -1 \\ 7 & -2 & 0 \\ 9 & -5 & -[4a] + 5 \end{vmatrix} = 0$$

$$\Rightarrow -1(-17) + (-[4a] + 5)(17) = 0$$

$$\Rightarrow 17 + 17(-[4a] + 5) = 0$$

$$\therefore [4a] = 6$$

$$\text{or } 6 \leq 4a < 7$$

$$\Rightarrow \frac{3}{2} \leq a < \frac{7}{4}$$

$$\therefore a \in \left[\frac{3}{2}, \frac{7}{4} \right)$$

49. Applying $R_3 \rightarrow R_3 + 2R_2$ and taking 20 common from R_3 then

$$\begin{vmatrix} f(x-2) & f(x+5) & f(x-2) \\ 5 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Now applying $C_2 \rightarrow C_2 - C_1$ and

$C_3 \rightarrow C_3 - C_1$, then

$$\begin{vmatrix} f(x-2) & f(x+5) - f(x-2) & 0 \\ \vdots & & \\ 5 & -1 & -10 \\ \vdots & & \\ 1 & 0 & 0 \end{vmatrix}$$

$$\Rightarrow -10\{f(x+5) - f(x-2)\} = 0$$

$$f(x+5) = f(x-2)$$

Replacing x by $x+2$, then

$$f(x+7) = f(x)$$

$$\therefore \text{Period} = 7$$

$$50. \quad |\text{adj}(\text{adj}A)| = |A|^{(n-1)^2} = |A|^9$$

$$(n-1)^2 = 9$$

$$\Rightarrow n-1 = \pm 3 \quad ; \quad n = 4$$

$$51. \quad \text{Put } x = 0, a = \Delta = 0, \text{ for } b$$

$$b = \Delta'(0), \Delta'(x) = b + 2cx$$

$$\Delta'(0) = b = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix} = -1$$

$$a - b = 1$$

$$52. \quad \frac{D}{(10!)^3} - 4$$

$$(10)(2 \cdot 10^2 + 8n + 10)$$

$$\text{Hence } \frac{D}{(10!)^3} - 4 = 2900$$